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Delineating spatiotemporal variations of coastal ecosystem service value from the 'past-current-future' perspective

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Abstract

Insufficient understanding of coastal natural resources hinders the efforts to reconcile economic development and ecological protection. The dynamics of ecosystem service value (ESV) needs to be assessed for opening opportunities in understanding the ecological consequences of human activities. This study identifies the spatiotemporal patterns of ESV from the "past-present-future" perspective, coupling hotspot analysis, geo-detector and patch-generating land simulation model into the framework of ESV assessment. Land use resources in Liaoning coastal zone over past two decades that are utilized for wetlands (-31.86%), croplands (-11.38%) and grasslands (-4.76%) decreased to different degrees, whereas built-up lands increased by 100.02%. The magnitude of coastal ESVs is 149.41 billion CNY over past two decades average, and has shown constant fluctuations as a consequence of land use changes. The distribution of ESVs evolved towards greater coverage of cold spots and smaller hot spot areas during 2000–2020 year. Watersheds and wetlands with 13.25% and 3.77% of total area contribute 70.37% and 8.29% of total ESVs, respectively, especially in water supply, hydrological regulation, and biodiversity. Compared to 2020 year, the value of ESVs in 2050 year increased to 212.72 and 311.48 billion CNY for economic and ecological (maximize ecosystem service/ecological capacity) scenarios, respectively, whereas decreased to 89.14 billion CNY for baseline scenario. Intense human activities are going to reshape the patterns of ESV generated from coastal natural resources. These findings provide timely and precisely evidence for balancing ecological protection and economic development in the context of Liaoning coastal construction from now on.

Highlights

1. Wetlands, croplands and grasslands decrease to different extents during 2000–2020 year, whereas built-up lands increases.
2. Ecosystem service value (ESV) is fluctuating due to the changes in watersheds and wetlands.
3. Future ESVs at 2050 year will be driven by land planning for divergent development scenarios.
4. Human activities are reshaping the pattern of coastal ESVs generated from natural resources.

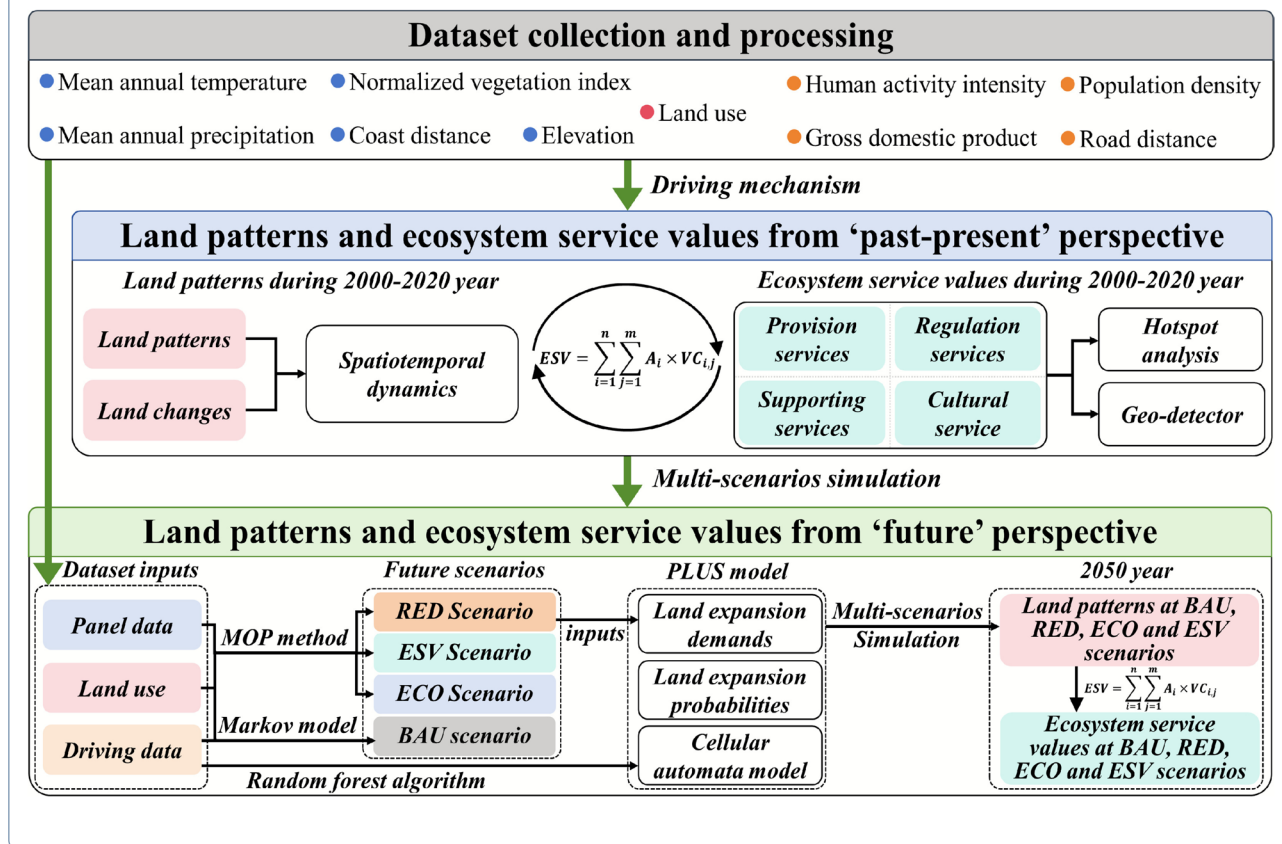
Keywords Land use changes, Liaoning coastal zone, Ecosystem service value, Driving factors, Multi-scenarios simulation

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Graphical Abstract



1 Introduction

Land use changes, being an essential performance of human activities, has been reshaping the structure and function of terrestrial ecosystems (Chillo et al. 2018; Felipe-Lucia et al. 2020). Uneven changes of land use pattern may potentially restrict the abilities of ecosystems to provide benefits for humans. The aggravative human activities (e.g., frequent land use changes) is reshaping ecosystem structure, with devastating tolls on the functions and services of regional ecosystem (Liu et al. 2024a; Mendoza-González et al. 2012; Zhang et al. 2020). More attentions should be paid to coastal natural resource, in comparison to the frequently noticed inland regions (Pablo et al. 2021; Yang et al. 2019; Zhang et al. 2024a), since the patterns of ESV (particularly water-related and regulating ecosystem services) in coastal zones are characterized by strong terrestrial-oceanic interactions and their responses to land-use change fundamentally different from those of non-coastal areas (Khamis et al. 2017; Ramesh et al. 2015; Wang et al. 2020). Nevertheless, in the context of flourishing ecosystem service value (ESV) assessment, the evidence for coastal zone is still sparse

(Costanza et al. 2017; Hasan et al. 2020), restricting those efforts to understand the ecological consequences of the coastal activities. Identifying the spatiotemporal variations of ESV in coastal zone provides insights into the coupling links from ecological functions to coastal development, thus paving the ways for achieving the sustainability of both civil society and planet.

Ecosystem services are those functions and processes related to the human benefits derived from ecosystems, either-directly and/or indirectly, including provisioning, regulating, supporting and cultural services (Balasubramanian 2019; Costanza et al. 1997; La Notte et al. 2017). As the regions of land-sea interaction, quantifying the magnitude of ESV in coastal zone is essential for the sustainable management of coastal natural resource (Liu et al. 2024a; Uehara et al. 2020). Existing methods for estimating ESVs include benefit transfer estimation, functional capacity valuation and the equivalent factor method (Jiang et al. 2021; Ma and Yang 2025), in particular, the equivalent factor method is intuitive and feasibility. One of the first attempts to assess the unit ESVs of diverse land use types was done by Costanza et al.

(1997). Afterwards, Xie et al. (2015) proposed a pricing framework for ecosystem services in China by taking into account both ecosystem and socio-economic conditions. The equivalent factor method has been widely adopted, especially in exploring the variations of ESV under land use changes. Along with the advent of Anthropocene, the dramatic alterations in land use patterns and ecosystem functions have been observed (Fu et al. 2022). The fluctuating ESVs in response to land use changes may open opportunities towards better understanding about the performance of ecosystem in the context of Anthropocene. There are numerous reports either exploring the variations of ESV across various scales (e.g., provinces, urban agglomerations, basins, and ecological zone) (Jiang et al. 2021; Ke et al. 2024; Malinga et al. 2015; Xu et al. 2024), or revealing the dynamics of ESV for specific ecosystem types (e.g., forests, grasslands and wetlands) (Acharya et al. 2019; Xu et al. 2020; Zhao et al. 2020b), largely focusing only on the patterns of ESV from the “past-present” perspective. These efforts towards recording the historical trends of ESVs may be somewhat outdated, considering that ecosystem in coastal zone is highly variable (Bevilacqua et al. 2015), and this is hindering accurately predict both current and future trends of regional ecosystems services. It is urgent to more deeply identify the dynamics of ESV in the context of diverse future scenarios.

The gross domestic product of coastal economic belt in Liaoning is 1,200 billion CNY at 2020 year, accounting for 48.9% of the total province (Zhang and Lu 2019). The coastal region has become a new engine for marine economic development of Liaoning province (Zhao et al. 2020a). Unfortunately, the coastal zone in Liaoning province was exposed to the risks from chaotic development, coast pollution, and dwindling fishery resources along with the deepening of marine development (Hou et al. 2024; Liu et al. 2013; Zhang et al. 2020). Against this background, rapid land-use conversion and urban expansion driven by human activities have become the dominant forces reshaping coastal landscapes and ecosystem functions over recent decades. Consequently, optimizing land-use patterns to protect ecosystem functions has emerged as a pressing issue for the ecologically sustainable development of the Liaoning coastal zone (Liu et al. 2024a, 2013; Uehara et al. 2020). It is beyond doubt that identifying the spatiotemporal patterns of ESV under land use change in Liaoning coastal zone can provide powerful evidence for assessing the ecological consequences of coastal development activities, since ESV is a valid indicator of ecosystem services (Luan et al. 2024; Qi et al. 2020). More importantly, many existing studies adopting a “past-present-future” framework are largely inland-focused, and their conclusions cannot be directly

transferred to coastal zones (Cen and Zhang 2024; Qiu et al. 2020; Schirpke et al. 2023); consequently, coastal ESV assessments integrating both historical dynamics and future land-use scenarios remain limited, constraining effective spatial optimization for long-term ecological security.

In view of this, this study is based on the “past-present-future” perspective to identify the spatiotemporal variations of ESVs in Liaoning coastal zone, by including multi-scenarios simulation, hotspot analysis, and geo-detector into the assessment framework of ESV. Three specific objectives of this study were to: (1) identify the spatiotemporal patterns of ESVs under land use changes in Liaoning coastal zone; (2) predict the patterns of land use and ESVs under diverse future scenarios; and (3) quantify the relative contribution of social-economic influences and ecological factors to the variability of coastal ESVs. The understanding about the long-term trends of coastal ESVs under land use change can lay the foundation for regional ecological protection and land management, and contribute to realizing the sustainable development of Liaoning coastal zone.

2 Material and methods

2.1 Study area and data sources

Liaoning province is the northernmost coastal province of China, located in the mid-latitude region (38°43'N~43°26'N, 118°53'E~125°46'E), belonging to the warm-temperate humid-semi-humid monsoon climate zone. The coastline of Liaoning province extends from the mouth of the Yalu River in the east to the political boundary of the Liaoning-Hebei Sea area in the west, with a total length of 2,737.6 km (2,110 km on the mainland and 627.6 km on the islands) (Figure S1).

The land use data of this study were derived from global land cover dataset of National Center for Basic Geographic Information (GlobeLand30). The coastal zone of Liaoning province, spanning a 10km range, was obtained through the buffer zone analysis. Subsequently, the mask extraction was employed to extract the land use change in Liaoning coastal zone. Combined with the division of terrestrial ecosystems by Xie et al. (2015) and land characteristics of Liaoning coastal zone, the land use data were reclassified into seven major types, including croplands, forests, grasslands, wetlands, watersheds, bare lands and built-up lands.

The determinants in this study include social-economic factors (e.g., human activity intensity, population density, gross domestic product and road distance) and ecological drivers [e.g., mean annual temperature and precipitation, normalized vegetation index (NDVI), natural topography (DEM) and coast distance]. Population density, gross domestic product, mean annual

temperature, mean annual precipitation, and normalized vegetation index were obtained from Resources and Environmental Science and Data Center, Chinese Academy of Sciences (CRES; <https://www.resdc.cn/>). The human activity intensity was calculated using the Delphi method (Wang and Zhao 2023; Zhang et al. 2024b), in which the conversion parameters of human activity intensity were 0.65, 0.11, 0.08, 0.16, 0.24, 0.95, 0.04 for croplands, forests, grasslands, wetlands, watersheds, built-up lands and bare lands, respectively. The distances to roads and coasts were calculated on the basis of OpenStreetMap dataset using Euclidean distance analysis. The dataset of natural topography (DEM) is derived from NASA SRTM1 V3.0. All spatial datasets were resampled to 30 m resolution and converted to the same projection (WGS_1984_UTM_Zone_51N). Finally, this study performed the spatiotemporal assessment of ESVs by inputting all datasets into a cascading framework, in which equivalent factor method, hotspot analysis, geo-detector and multi-scenarios prediction are incorporated from a “past-present-future” perspective (Fig. 1).

2.2 The assessment of ecosystem service value

The economic value per unit of ecosystem service was corrected based on the equivalent factor table proposed by Xie et al. (2015) and combined with the actual statistics of grain cultivation area, yield, and unit price in the study area. To ensure temporal stability and representativeness, the grain statistics (sown area, yield, and unit price) used in Eq. (1) were calculated as multi-year averages over the period 2000–2020. The calculation method is as follows:

$$E_a = \frac{1}{7} \sum_{j=1}^n \frac{m_j p_j q_j}{M} \quad (1)$$

where E_a is the economic value of per unit of ecosystem service in CNY/(hm² year); j is the types of grain crops; n is the number of grain crops types; m_j is the average economic value of the j th grain crops in study area during 2000–2020 (CNY/kg); p_j is the yield of the j th grain crops in study area during 2000–2020 [kg/(hm² year)]; and q_j is the planted area of the j th grain crop in study area during 2000–2020 (hm²); M is total planting area of grain crops in study area during 2000–2020 (hm²). The revised economic value per unit of ecosystem service in

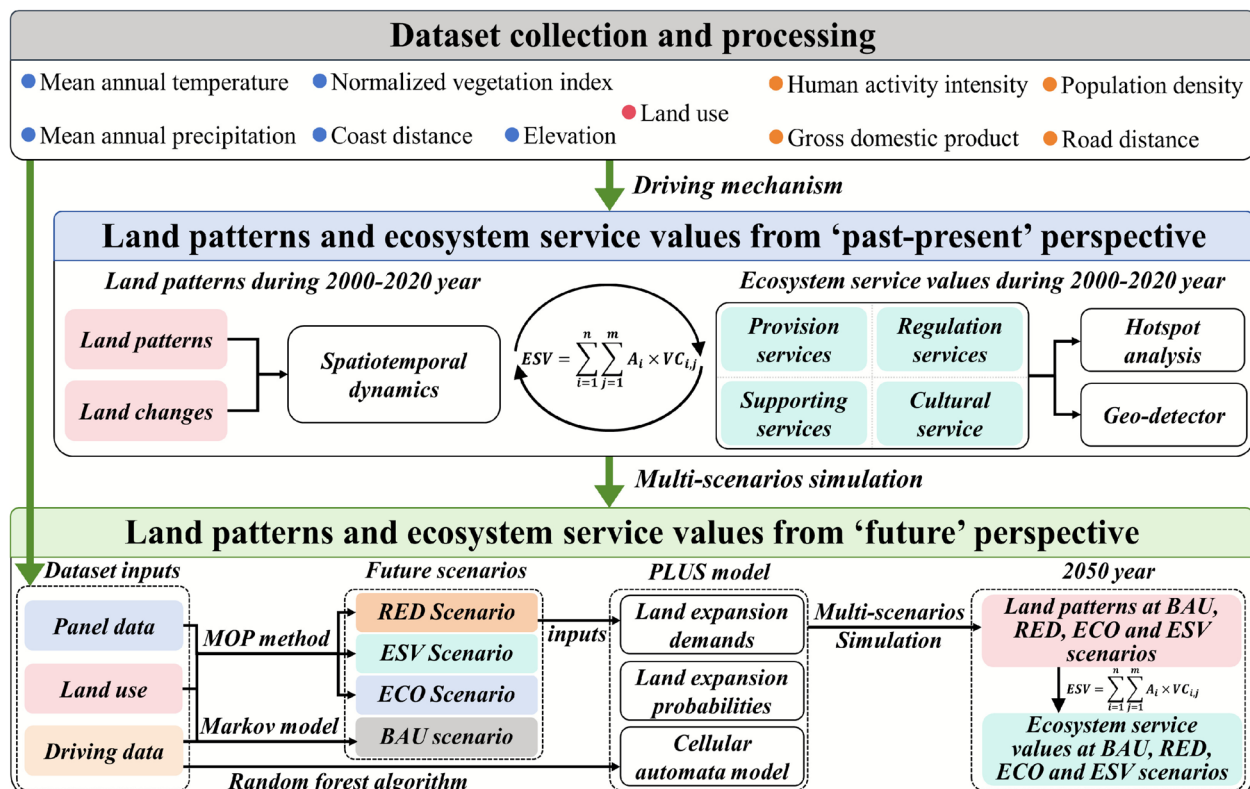


Fig. 1 Depicts the cascade framework for ecosystem service value assessment from the ‘past-current-future’ perspective. (In which MOP denotes multi-objective programming approach, PLUS model is patch-generating land simulation model, BAU, RED, ECO and ESV scenarios are baseline, economic, ecological capacity and ecosystem service scenarios, respectively)

Liaoning coastal zone was 3933.15 CNY/(hm² year). The actual equivalent factors table was obtained on the basis of revised economic value and the table of equivalent factors (Table 1).

The formula for calculating the ESV in the study area is as follows:

$$ESV = \sum_{i=1}^n \sum_{j=1}^m A_i \times VC_{i,j} \quad (2)$$

where *ESV* is the magnitude of ESV in CNY/year; *n* is the number of land use types; *m* is the number of ecosystem service types; *A_i* is the area of the *i*_{th} land use type in the study area in hm²; and *VC_{i,j}* is the equivalent value of ecosystem service of the *j*_{th} ecosystem service of the *i*_{th} land use type in CNY/(hm² year).

2.3 Spatial pattern analysis

The index of global Moran's *I* was used to test the spatial autocorrelation of ESVs in Liaoning coastal zone at grid scale. Moreover, hotspot analysis also was employed to detect the spatial dynamics of ESVs in Liaoning coastal zone at grid scale. In this study, "hot spots" refer to spatial clusters where grid cells with high ESVs are surrounded by neighboring cells that also exhibit high ESVs, indicating statistically significant high-value aggregation. Conversely, "cold spots" denote clusters of grid cells with consistently low ESVs relative to their surrounding areas, reflecting statistically significant low-value aggregation. The specific formula is defined as follows:

$$G_i^* = \frac{\sum_{j=1}^n w_{ij}x_j - X \sum_{j=1}^n w_{ij}}{\sqrt{\frac{[n \sum_{j=1}^n w_{ij}^2 - (\sum_{j=1}^n w_{ij})^2]}{(n-1)}}} \quad (3)$$

$$X = \frac{1}{n} \sum_{i=1}^n x_i \quad (4)$$

$$S = \sqrt{\frac{1}{n} \sum_{j=1}^n x_j^2 - (X)^2} \quad (5)$$

where *G_i^{*}*, *X* and *S* are the values of z-score, mean and standard deviations of ESVs at specific grid, and higher values of *G_i^{*}* indicates that high-value clustering (hot spots) of ESVs is closer, and vice versa. *n* is the number of spatial grid units; *x_i* and *x_j* are the observed values of the *i*_{th} and *j*_{th} grid unit, respectively; and *w_{i,j}* is a spatial weight matrix based on spatial adjacency.

2.4 Future scenarios of land use pattern

A multi-objective programming, which can integrate various ecological resources and macroeconomic policies, was employed to set up diverse future scenarios of land use changes through appropriately defined optimization objective and constraints conditions. In this study, three optimization targets are preset with the objective function and constraint function (Tables 2 and 3), including (1) maximize economic benefits; (2) maximize ecosystem service; and (3) maximize ecological capacity. The structure and parameterization of the objective functions were primarily adapted from previous multi-objective land-use optimization studies (Liang et al. 2021), which provide a well-established framework for balancing economic development and ecological protection. To ensure model robustness and comparability with existing studies, most coefficients in the objective functions were directly inherited from the literature, only the coefficients associated with wetlands were slightly adjusted to better reflect the coastal zones context, where wetlands play a disproportionately important role in ecosystem service provision and ecological regulation.

Table 1 The actual equivalent factors table of ecosystem service value per unit area in Liaoning coastal zone [CNY/(hm² year)]

Category I	Category II	Croplands	Forests	Grasslands	Wetlands	Watersheds
Providing services	Food product	3343.24	1140.63	1494.62	2005.94	3146.58
	Material product	1573.29	2595.93	2202.61	1966.61	904.64
	Water supply	78.66	1337.3	1219.3	10,187.05	32,606.42
Regulating services	Gas treatment	2635.26	8535.09	7748.45	7473.12	3028.58
	Climate regulate	1415.96	25,565.95	20,492.09	14,159.6	9007.08
	Waste treatment	393.32	7591.12	6765.14	14,159.6	21,829.39
	Hydrological regulate	1061.97	18,643.48	15,024.91	95,302	402,132.75
Supporting services	Soil retention	4051.22	10,423.04	9439.74	9085.75	3657.9
	Nutrient cyclings	471.99	786.64	707.98	707.98	275.33
	Biodiversity	511.32	9479.07	9479.07	30,954.47	10,029.72
Cultural service	Landscape aesthetics	235.99	4169.22	3775.89	18,604.15	7433.79

The actual ESV per unit area of bare and built-up lands is zero

Table 2 The optimization targets for multi-scenarios of multi-objective programming in 2050 year at Liaoning coastal zone

Multi-scenarios	Objective function
Economic scenario	$F_{(x)} = \max (6.70x_1 + 1.88x_2 + 198.04x_3 + 2.85x_4 + 1.69x_5 + 1831.26x_6 + 0.00x_7)$
Ecosystem service scenario	$F_{(x)} = \max (5.80x_1 + 20.63x_2 + 8.56x_3 + 20.86x_4 + 33.27x_5 + 0.00x_6 + 1.02x_7)$
Ecological capacity scenario	$F_{(x)} = \max (5.00x_1 + 1.76x_2 + 0.08x_3 + 10.54x_4 + 9.42x_5 + 2.50x_6 + 0.00x_7)$

$x_1 \sim x_7$ represent the area (ha) of croplands (x_1), forests (x_2), grasslands (x_3), wetlands (x_4), watersheds (x_5), built-up lands (x_6), and bare lands (x_7), respectively

Table 3 The constraint functions for multi-scenarios of multi-objective optimization programming in 2050 year at Liaoning coastal zone

Constraint function	Description
$\sum_{i=1}^7 x_i = 2694432$	Ensures land-area conservation (the total area of all land-use types is fixed)
$60\% \leq \frac{x_1}{x_6} \leq 80\%$	Constrains the cropland-built-up land balance to avoid excessive cropland loss under urban expansion and to maintain basic food-production capacity, and we assume that the ratio of cropland to built-up land should be between 60.00% and 80.00%
$2.50\% \leq \frac{x_3 + x_7}{2694432}$	Maintains minimum grassland and bare land share to preserve land-use diversity and provide flexibility for land-structure adjustment, it is assumed that bare land and grassland should occupy 2.50% of total land area
$22\% \leq \frac{0.49 \cdot x_3 + x_2 + 0.46 \cdot x_1}{2694432}$	"Green equivalent" constraint (weighted greening contribution) to ensure baseline ecological greenness/vegetation structure, following the commonly used greening-coefficient approach (Liu et al. 2002). The total green equivalent is assumed to be greater than 22.00% by 2050
$45\% \leq \frac{x_4}{x_5} \leq 65\%$	Controls the wetland-watershed structure to maintain coastal ecological quality and water-related regulating functions, wetland and watershed should meet certain proportions, ranging from 45.00% to 65.00%
$1.12\% \leq \frac{x_4}{2694432}$	Sets a minimum wetland proportion (1.12%) to safeguard coastal biodiversity and key regulating services
$23.50\% \leq \frac{x_6}{2694432} \leq 35.50\%$	Keeps built-up land within a reasonable urbanization range consistent with sustainable development, the proportion of built-up land should be maintained within a certain range, 23.50% to 35.50%
$25.00\% \leq \frac{x_5}{2694432} \leq 37.50\%$	Ensures a reasonable watershed proportion (25.00 ~ 37.50%) to support water supply and hydrological regulation in the coastal zone
$1.35\% \leq \frac{x_7}{2694432} \leq 3.50\%$	Maintains bare land within a realistic range (1.35 ~ 3.50%) and preserves structural adjustment space
$32.50\% \leq \frac{x_7}{x_2} \leq 65.50\%$	Prevents extreme land-structure solutions by constraining the bare land-forest relationship, the ratio of bare land to forest should be between 32.50% and 65.50%

Constraint thresholds represent planning-oriented bounds commonly used in multi-objective land-use optimization

Constraint conditions (Table 3) were specified to ensure that the optimized land-use demands are both feasible and consistent with the land-resource endowment and coastal management requirements of the Liaoning coastal zone. Following the constraint-setting strategy commonly adopted in multi-objective land-use optimization studies (Liang et al. 2021), most constraints and thresholds were inherited from existing frameworks that represent planning-oriented boundaries that reflect widely used assumptions in land-use optimization and scenario analysis.

We predicted the land use demand of Liaoning coastal zone under three future scenarios by solving three optimization objective functions using Lingo 12.0 software. Furthermore, a baseline scenario, which stands for the empirical trend of land use in study area,

was also derived through Markov chain approach. We were setting 2050year as the target year for the future.

2.5 Patch-generating land simulation model

The predicted demands for land use under four future scenarios were input into the patch-generating land simulation model to further simulate the land pattern of in Liaoning coastal zone at 2050year (Liang et al. 2021). For details, we first explored the role of social-economic factors and ecological drivers in driving the expansion of each land use type using random forest algorithm. Subsequently, the analysis of land expansion strategy was further adopted to examine land expansion pattern of land use growth spanning two decades for extracting the expansion probabilities of each land type. Ultimately, in the constraints of land demands and expansion probabilities, a cellular automata model

combining random seed generation and threshold decreasing mechanism is constructed to simulate land pattern of Liaoning coastal zone under four future scenarios.

2.6 Geo-detector

Geo-detector is a statistical tool for detecting the relative contributions of driving factors to spatial heterogeneity (Wang et al. 2016). In this study, the detector of factor and interaction in geo-detector were employed to diagnose the dominant factors and their interactions affecting the spatial heterogeneity of ESVs in Liaoning coastal zone. The expression of the detector is as follows:

$$Q = 1 - \frac{\sum_{h=1}^1 N_h \sigma_h^2}{N \sigma^2} \quad (6)$$

where Q is the explanation degree of driving factor to spatial heterogeneity of ESVs; h is the stratification of variable; σ_h^2 and σ^2 represent the variance between the layer

sample and the whole sample; N_h and N are the layer h and the number of units in the whole layer.

3 Results

3.1 Land pattern and ecosystem service value from the 'past-present' perspective

It was observed that the area of croplands accounted for 49.24% ($684,668.64 \pm 34,280.95 \text{ hm}^2$) of Liaoning coastal zone during the period of 2000–2020 year, followed by built-up lands ($227,821.05 \pm 65,456.27 \text{ hm}^2$; 16.38%) and watersheds ($184,327.32 \pm 3,269.90 \text{ hm}^2$; 13.25%), respectively (Fig. 2). Forests and grasslands cover 9.12% ($126,800.52 \pm 14,820.18 \text{ hm}^2$) and 4.93% ($68,549.43 \pm 11,145.58 \text{ hm}^2$) of the total area of Liaoning coastal zone, respectively (Fig. 2).

There was a notable expansion of built-up lands, grasslands and watersheds over the period of 2000–2010 year, encroaching on considerable areas of croplands and bare lands. The transformation of land use has been mainly observed in the conversion of bare lands to watersheds ($24,287.33 \text{ hm}^2$), croplands to built-up lands ($14,163.10$

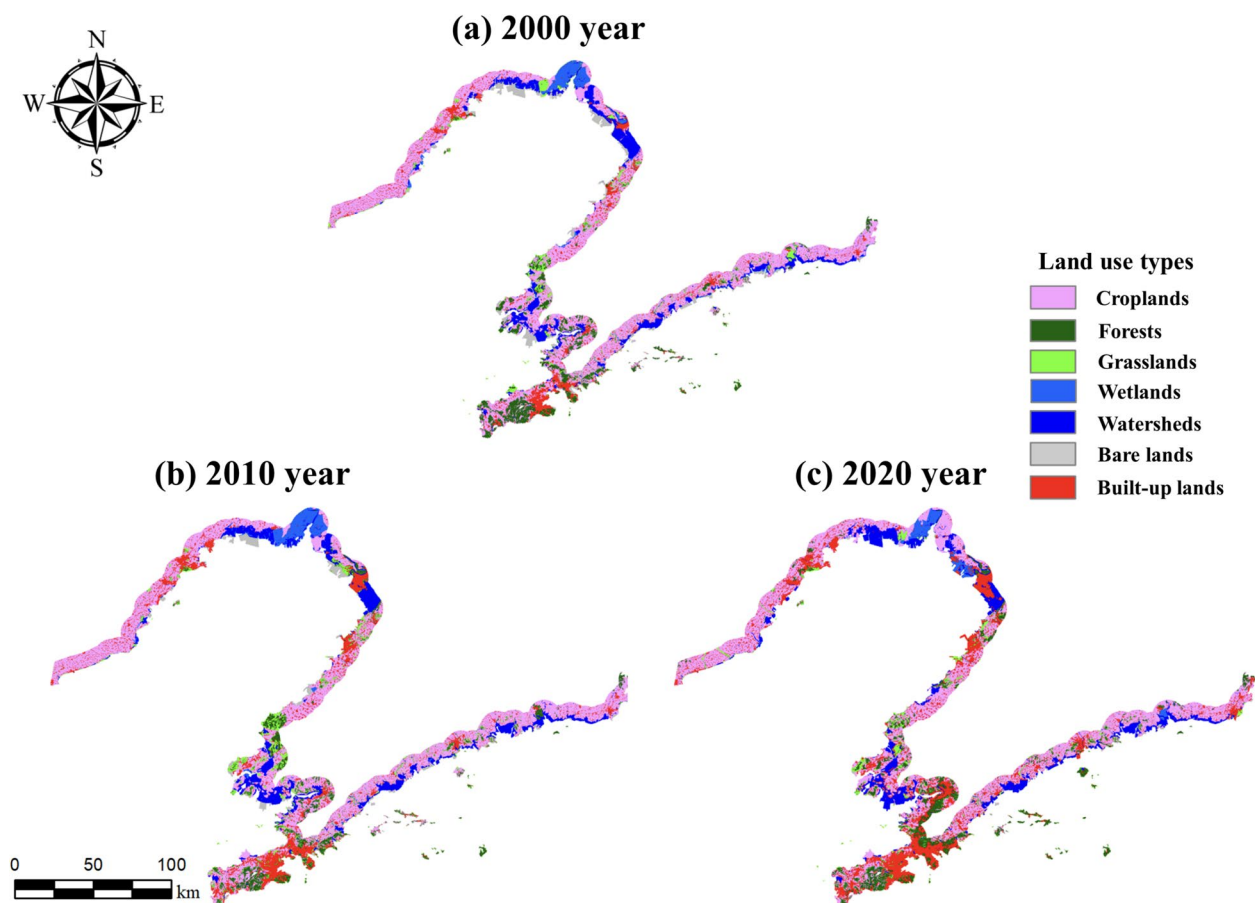


Fig. 2 The spatial patterns of land use within Liaoning coastal zone at (a) 2000; (b) 2010; and (c) 2020 year. Each colored branches represent specific land use type, with croplands, forests, grasslands, wetlands, watersheds, bare lands and built-up lands

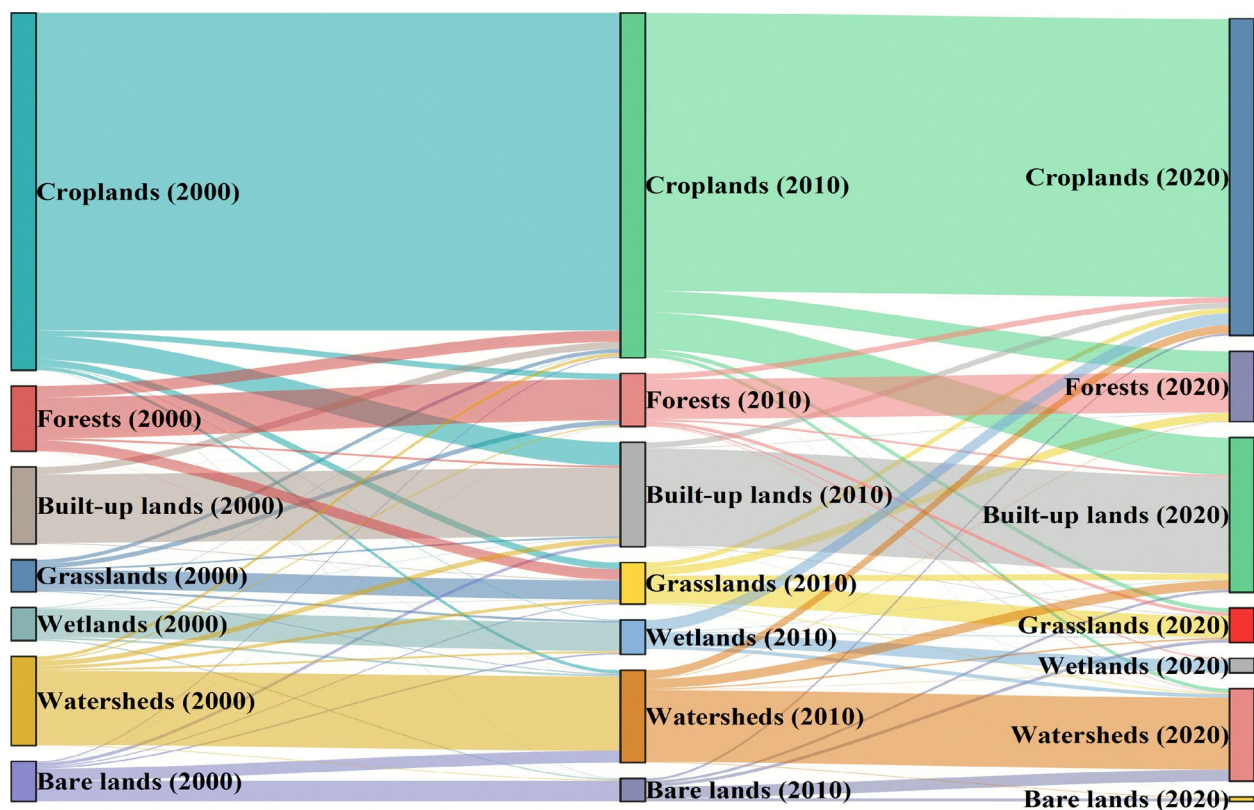


Fig. 3 The transformation of land use in Liaoning coastal zone from 2000 to 2020 year. Each colored branch represents specific land use type, with croplands, forests, built-up lands, grasslands, wetlands, watersheds and bare lands

hm²), forests to grasslands (21,450.52 hm²) and croplands (23,182.56 hm²), respectively, in the first decade (Fig. 3). In contrast, during 2010–2020 year, the extensive areas of croplands, wetlands and watersheds have been reclaimed for built-up lands, forests and grasslands, in particular, with the expansion of built-up lands and forests taking up croplands. The land expansion in the second decade is expressed in the conversion of croplands to built-up lands (74,933.90 hm²) and forests (42,796.58 hm²), wetlands to croplands (24,060.16 hm²), watersheds to built-up lands (18,892.70 hm²) and croplands (16,445.69 hm²), and bare lands to watersheds (23,311.73 hm²) (Fig. 3).

The transformation of land use is profoundly influencing the structure and functionality of ecosystems, bringing about the alterations in ESVs magnitude. The amount of ESVs in Liaoning coastal zone has increased from 148.73 to 151.91 billion CNY in the first decade, but decreased by 4.31 billion CNY over the second decade (Table 4). The fluctuations of ESVs are primarily in hydrological regulation, biodiversity and landscape aesthetics over the two decades, which can be attributed to the changes of watersheds and wetlands. For instance, in the period of 2000–2020 year, the area of watersheds trended upwards (from 179,704.53 hm² to 186,534.99 hm²),

Table 4 The magnitude of ecosystem service value in Liaoning coastal zone during 2000–2020 year

Primary category	Ecosystem service value (billion CNY)		
	2000 year	2010 year	2020 year
Providing services	13.84	13.99	13.40
Regulating services	116.62	119.69	117.30
Supporting services	14.38	14.29	13.33
Cultural service	3.90	3.94	3.57
Total ecosystem service value	148.73	151.91	147.60

bringing about the additional hydrological regulation of 2.75 billion CNY. The amount of hydrological regulation (changed from 6.42 to 6.51 and then to 4.37 billion CNY) and biodiversity (changing from 2.09 to 2.11 and then to 1.42 billion CNY) derived from wetlands also fluctuated with the changes of wetlands (Fig. 4). Notably, the magnitude of ESVs provided by each land use type also diverges. To be more specific, watersheds contributed the majority amount of ESVs, providing 70.37% of total ESVs within coastal zone with the average area of 13.25% over

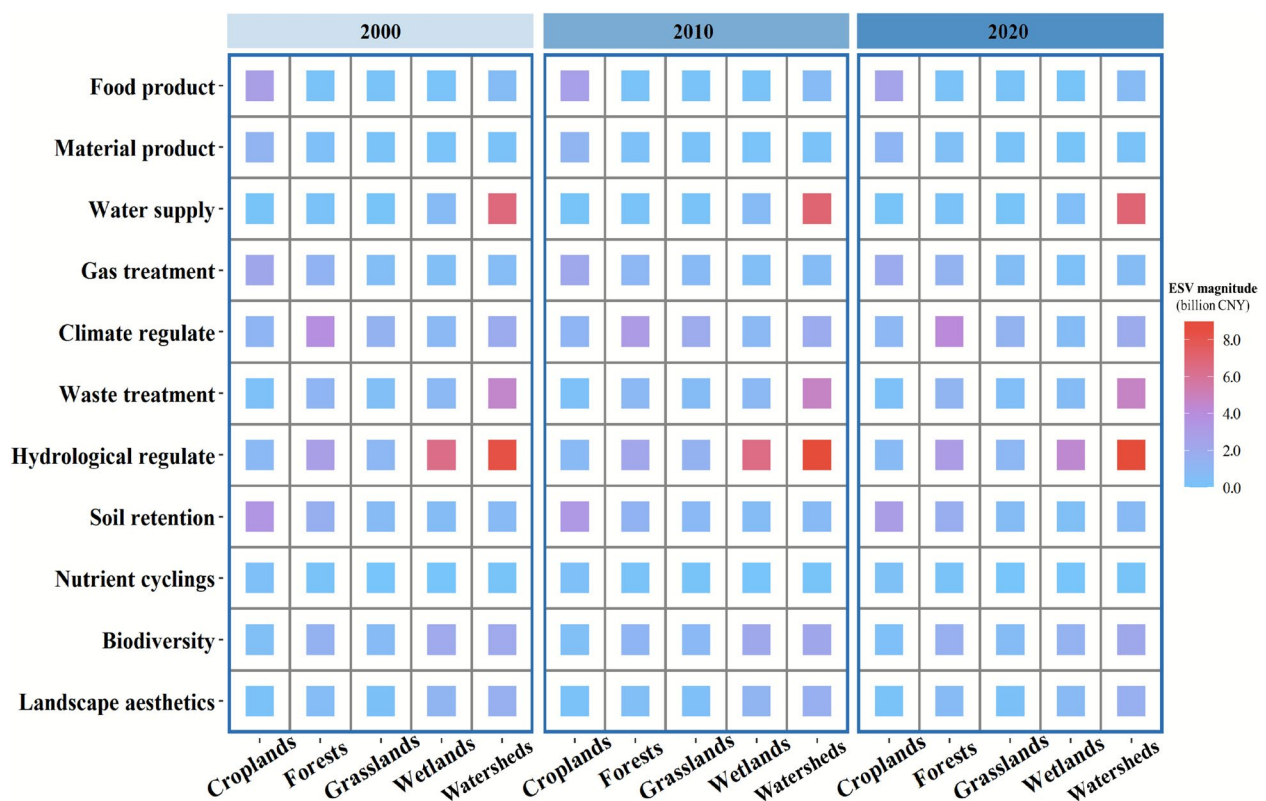


Fig. 4 The magnitude of ecosystem service value for each service types of each land types in Liaoning coastal zone during 2000–2020 year.

The elements in the matrix represent the magnitude of the corresponding ecosystem service value provided by different land use types, including croplands, forests, grasslands, wetlands and watersheds. Food product, material product and water supply belong to providing services, gas treatment, climate regulate, waste treatment and hydrological regulate belong to regulating services. Supporting services include soil retention, nutrient cycling and biodiversity. Cultural service focuses on landscape aesthetics

the past two decades (Figs. 2 and 4). The magnitude of ESVs for forests (126,800.52 hm², 9.12%) and croplands (684,694.70 hm², 49.24%) equaled 13.22 billion CNY and 12.37 billion CNY, respectively, with contributions to 8.85% and 8.28% of total ESVs (Figs. 2 and 4). Grasslands and wetlands covered 4.93% and 3.77% of total area, and provided 6.20 and 12.38 billion CNY in services to human benefits (Figs. 2 and 4).

The spatial pattern of ESVs in Liaoning coastal zone is characterized by interspersed hot and cold spots (Fig. 5). The cold spots occurred principally in Huludao, Yingkou and Dalian, with large areas of croplands and built-up lands of low level of ESVs (Fig. 2). The hot regions with higher ESVs are mainly distributed in four bands, including Jinzhou-Panjin-Yingkou, Wafangdian, Zhuanghe and Dandong, all of which contain a considerable area of wetlands and watersheds (Figs. 2 and 5). Especially, wetlands on the east coast of Liaohe Estuary underwent extensive reclamation for croplands over the past two decades, causing the decline of ESVs (Figure S2). The hot spots of ESVs in Liaoning coastal zone are progressively fragmented and sporadic along with the changes in land

patterns among 2000–2020 year (Fig. 5). The high-value clusters of ESVs gradually weakened and low-value clusters of ESVs kept expanding (Figure S2), which indicated the deterioration in environmental health and ecological functions of Liaoning coastal zone.

3.2 Land patterns and ecosystem service value under the ‘future’ perspective

The land patterns of Liaoning coastal zone in 2050 year were simulated for four future scenarios (e.g., baseline, economic, ecosystem service and ecological capacity scenarios) by incorporating multi-objective programming algorithm and patch-generating land simulation model. For baseline scenario, the area of built-up lands and croplands covered 521,950.71 hm² and 498,653.08 hm², respectively, followed by forests (186,053.33 hm²) and watersheds (119,141.80 hm²) (Table 5). Grasslands (56,494.07 hm²) in baseline scenario is found to be more than other scenarios, whereas the area of wetlands (5,827.50 hm²) appears smaller than other scenarios. The area of built-up lands (489,881.89 hm²), watersheds (347,909.51 hm²) and croplands (313,406.37 hm²) are

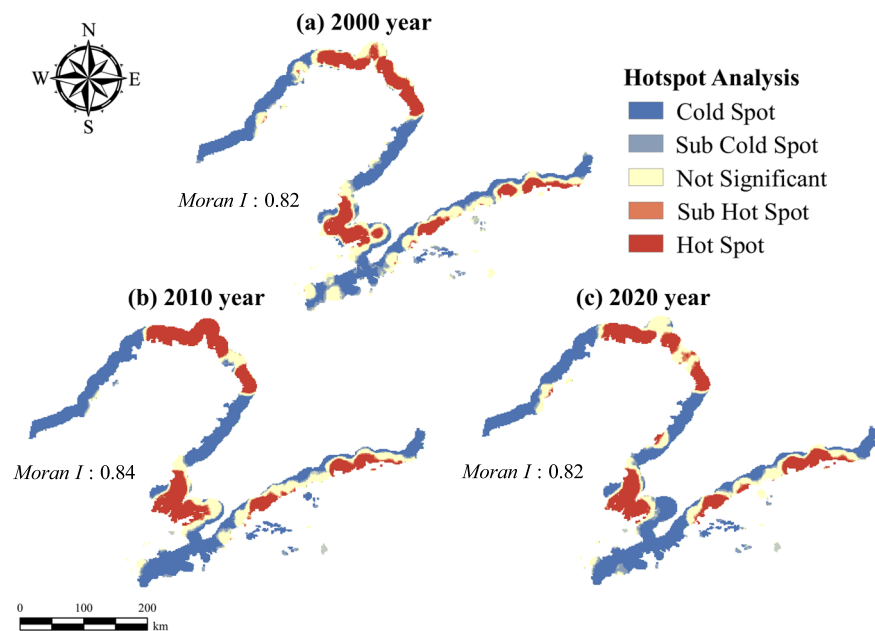


Fig. 5 Spatial patterns of hot spots of ecosystem service value in Liaoning coastal zone at (a) 2000; b 2010; and (c) 2020 year. Blue and red zone distinguish the cold spots and hot spots of ecosystem service value in Liaoning coastal zone

Table 5 The magnitude and proportion of land use demands in Liaoning coastal zone under four future scenarios in 2050 year

Land type	Baseline scenario		Economic scenario		Ecosystem service scenario		Ecological capacity scenario	
	Area/hm ²	Proportion/%	Area/hm ²	Proportion/%	Area/hm ²	Proportion/%	Area/hm ²	Proportion/%
Croplands	498,653.08	35.86	313,406.37	22.54	195,980.52	14.09	229,057.23	16.47
Forests	186,053.33	13.38	29,094.17	2.09	49,140.53	3.53	29,094.17	2.09
Grasslands	56,494.07	4.06	15,574.07	1.12	15,574.07	1.12	15,574.07	1.12
Wetlands	5827.50	0.42	156,559.20	11.26	233,399.47	16.78	233,399.47	16.78
Watersheds	119,141.80	8.57	347,909.51	25.02	518,665.90	37.30	518,665.90	37.30
Built-up lands	521,950.71	37.54	489,881.89	35.23	326,634.37	23.49	326,634.37	23.49
Bare lands	2404.90	0.17	38,100.17	2.74	51,130.52	3.68	38,100.17	2.74
Total	1,390,525.38	100.00	1,390,525.38	100.00	1,390,525.38	100.00	1,390,525.38	100.00

considered to be larger in economic scenario than in other scenarios. Wetlands (156,559.20 hm²) under economic scenario becomes smaller than 233,399.47 hm² for ecological (ecosystem service and ecological capacity) scenarios. Coastal land pattern under ecosystem service scenario is consistent with ecological capacity scenario. The areas of main land types, such as watersheds (518,665.90 hm²), built-up lands (326,634.37 hm²) and wetlands (233,399.47 hm²), are also equivalent in study area under ecosystem service and ecological capacity scenarios (Table 5).

The land use patterns of Liaoning coastal zone have not undergone obvious reforms (Fig. 6). The built-up lands

under four future scenarios mainly located in Dalian, Yingkou, and Huludao, while wetlands are concentrated in Panjin and Jinzhou and watersheds are clustered in Jinzhou, Yingkou and Dalian. Besides, forests and grasslands are dispersed sporadically in Dalian City. Compared with baseline scenario, the conversion of croplands alongside watersheds into built-up lands concentrated in Pulandian, Zhuanghe and Donggang under economic scenario. The area of watersheds in ecosystem service and ecological capacity scenarios are larger, reflecting in Linghai, Jinzhou, Wafangdian, Dalian and Zhuanghe (Fig. 6).

Compared with baseline scenario (89.14 billion CNY), the magnitude of ESVs in Liaoning coastal zone under

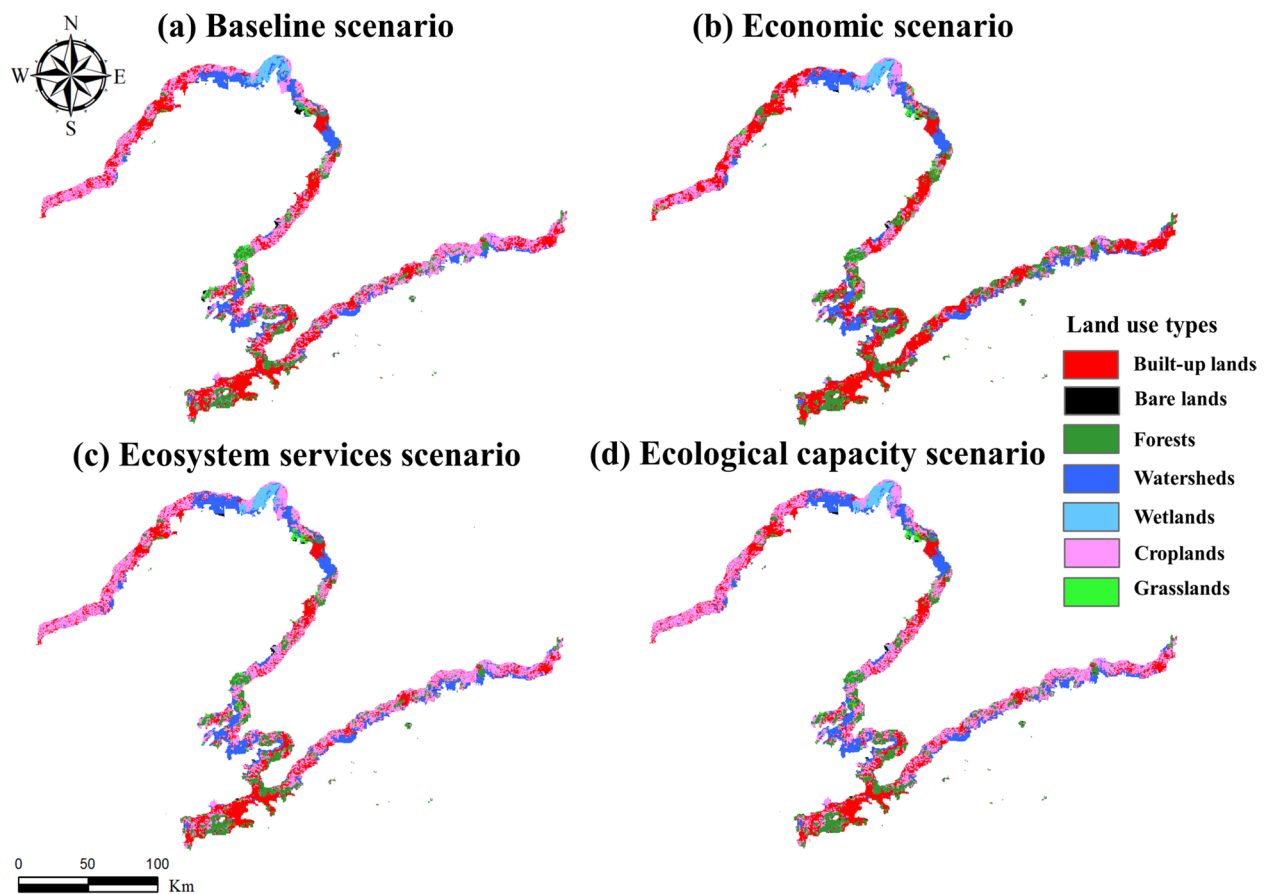


Fig. 6 Spatial pattern of land use in Liaoning coastal zone under baseline scenario (a), economic scenario (b), ecosystem service scenario (c) and ecological capacity scenario (d) at 2050 year. Each type of land use, including built-up lands, bare lands, forests, watersheds, wetlands, croplands and grasslands, are shown in different colors

economic scenario grows markedly up to 212.72 billion CNY, but remains lower than that of ecosystem service (312.78 billion CNY) and ecological capacity (311.48 billion CNY) scenarios (Table 6). The amount of ESV for all services, except for material production, fall below those of ecosystem service and ecological capacity scenarios, with the maximum gap recorded for the hydrological regulation (76 billion CNY) (Table 6). The gap for water supply, waste treatment, biodiversity, climate regulation and landscape aesthetics are 6.36, 4.86, 4.14, 2.74 and 2.72 billion CNY, respectively (Table 6). Moreover, the magnitude of most services in ecosystem service scenario exceeds ecological capacity scenario slightly, except for the minor deficit of 0.09 billion CNY for food production (Table 6).

The distinctions in spatial pattern of ESVs at Liaoning coastal zone under four future scenarios occurred mainly in the regions of Jinzhou, Panjin, Wafangdian, Pulandian and Zhuanghe (Fig. 7). The amount of ESV for economic scenario increased considerably in Huludao, Gaizhou, Dalian and Pulandian in contrast to baseline scenario.

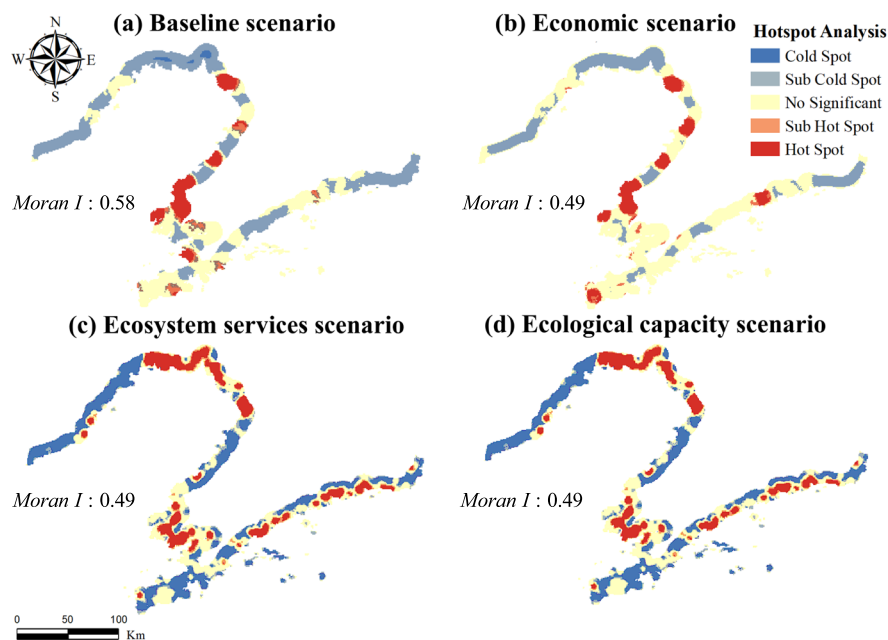
In both ecosystem service and ecological capacity scenarios, the regions with high ESVs tended to concentrate on Yingkou and Dalian, somewhat similar to the spatial pattern of ESV under economic scenario (Figs. 7 and S3). The gap of ESVs for ecological (maximize ecosystem service or ecological capacity) scenarios are observed in Jinzhou, Panjin, Yingkou and Wafangdian, Pulandian and Zhuanghe as compared to the economic scenario (Fig. 7), all of which presented higher ESVs magnitude under ecosystem service and ecological capacity scenarios than under economic scenario (Figure S3).

3.3 Driving factor of ecosystem service value of Liaoning coastal zone

Ecological factors and social-economic drivers jointly constrained the spatial pattern of ESVs in Liaoning coastal zone, the relative contribution of which is 62.13% and 37.87% respectively (Fig. 8). The Q-index of human activity intensity, as the main factor leading to the spatial characteristics of ESVs, was 0.42, followed by climatic conditions (0.30 and 0.21 for mean annual precipitation

Table 6 The magnitude of ecosystem service value for each service types in Liaoning coastal zone under four future scenarios in 2050 year

Primary category	Subcategories	Ecosystem service value/billion CNY			
		Baseline scenario	Economic scenario	Ecosystem service scenario	Ecological capacity scenario
Providing services	Food product	2.35	2.51	2.84	2.92
	Material product	1.51	1.223	1.4	1.4
	Water supply	4.3	13.02	19.39	19.37
Regulating services	Gas treatment	3.74	3.42	4.37	4.29
	Climate regulate	7.78	6.86	9.83	9.36
	Waste treatment	4.67	10.27	15.2	15.05
	Hydrological regulate	53.31	155.94	232.18	231.84
Supporting services	Soil retention	4.98	4.42	5.48	5.4
	Nutrient cycling	0.46	0.39	0.45	0.45
	Biodiversity	3.93	8.92	13.14	12.97
	Landscape aesthetics	2.1	5.75	8.51	8.43
Total ecosystem service value		89.14	212.72	312.78	311.48

**Fig. 7** Spatial distribution of ecosystem service value in Liaoning coastal zone under baseline scenario (a), economic scenario (b), ecosystem service scenario (c) and ecological capacity scenario (d) at 2050 year. Blue and red zone distinguish the cold spots and hot spots of ecosystem service value in Liaoning coastal zone

and temperature, respectively), natural topography (0.23 for DEM) and vegetation condition (0.15 for NDVI) (Fig. 8). Moreover, the Q-index of other social-economic factors (domestic product, population density and road distances are 0.05, 0.04 and 0.04, respectively) are less than 0.10, with little influence in ESVs (Fig. 8). It is also

worth pointing out that the interaction of all determinants exerted stronger effects on the spatial characteristics of ESVs than the contribution of singular drivers, suggesting that the interaction of the drivers is synergistic (Fig. 8). The magnitude of regional ESVs is diverse owing to various basic ecological conditions, forming the basic

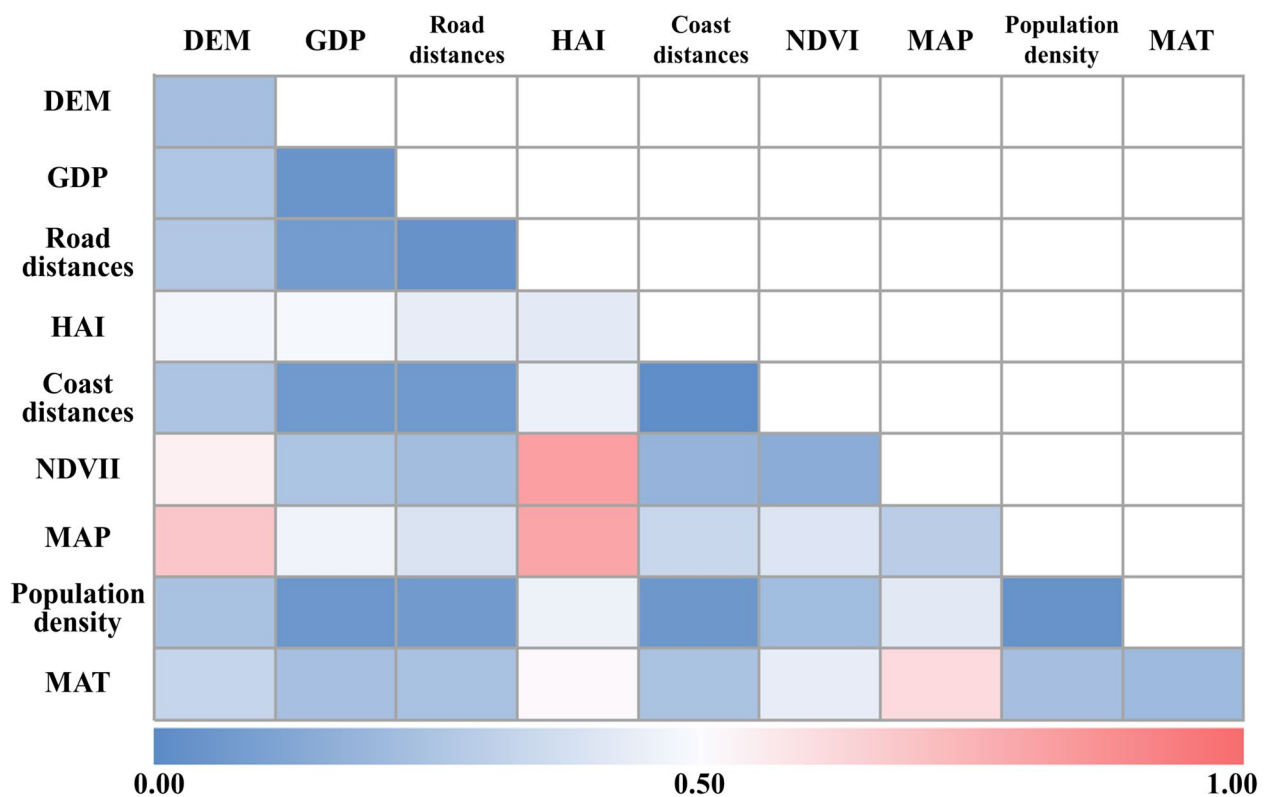


Fig. 8 Factor interaction detection of spatial distribution of ecosystem service value in Liaoning coastal zone. Color is the explanation degree of driving factor. DEM, HAI, NDVI, MAP and MAT are natural topography, human activity intensity, normalized vegetation index, mean annual precipitation and temperature

spatial pattern of coastal ESVs. Meanwhile, with human activities, the changes in land pattern further alter the spatial characteristics of ESVs in Liaoning coastal zone.

4 Discussion

4.1 Patterns of ecosystem service value and land expansion in Liaoning coastal zone

The continued urbanization in Liaoning coastal zone over past two decades has occupied enormous areas of ESV-rich lands (e.g., watersheds and wetlands), especially in high urbanized regions within Dalian, Yingkou and Huludao. The area of built-up lands in Liaoning coastal zone raised by 100.2%, of which, about 18,892.70 hectares comes from watersheds. Expanding built-up lands should be responsible for the fragmentation of continuous ecological space, that directly weakened the functionality of regional ecosystems (Ayinuer et al. 2024; Fu and Zhang 2014; Yuan et al. 2024). For instance, the fragmentation of watersheds and wetlands prevents both species movement and interaction, leading to the degradation of biodiversity (Felipe-Lucia et al. 2020; Nelson et al. 2009). The fragmentation of wetlands over the second decades contributed to a 31.8% decline in hydrological regulation and biodiversity of Liaoning coastal zone. Meanwhile, the

radical changes of land use might weaken the stability and resilience of regional ecosystems, thus further reducing the magnitude of ESVs (Chillo et al. 2018; Felipe-Lucia et al. 2020; Fu et al. 2015). Incidentally, in the context of croplands occupied by built-up lands, some of these watersheds and wetlands are also reclaimed for croplands for the purpose of food security, which also weakens the magnitude of ESVs of regional ecosystem.

These observed dynamics underscore the unique ecological vulnerability of coastal regions. Unlike inland areas, the ESV structure of the Liaoning coastal zone is characterized by a heavy dominance of regulating and water-related services (e.g., hydrological regulation and water supply) tied to transitional aquatic ecosystems (Khamis et al. 2017). While inland ESV patterns are often dominated by provisioning services or forest-related functions, coastal ESVs are far more sensitive to land conversions affecting the land-sea interface (Niu et al. 2024; Wang et al. 2014). Moreover, land-use changes here are driven by concentrated port development and maritime infrastructure, leading to sharper spatial trade-offs between economic growth and ecological integrity than those seen in the hinterlands (Wei et al. 2024; Wu et al. 2024). Such differences necessitate a comprehensive

understanding of the specific natural and anthropogenic forces that uniquely regulate the spatiotemporal pattern of coastal landscape for effectively managing these complex trade-offs.

The spatiotemporal patterns of coastal land use are regulated by both natural and anthropogenic drivers. The distribution of specific land type (e.g., wetlands and watersheds) with high levels of ESV is generally determined by natural forces like mean annual precipitation, topography and NDVI. The aggravated socio-economic activities, e.g., urbanization and road network expansion, frequently disrupt ecological zone, thereby contributing to the decline of regional ESVs (Li et al. 2010; Perumal et al. 2021). In spite of the fact that protective policies (e.g., wetlands restoration, afforestation and return of croplands to lakes) can partially constrain the uncontrolled expansion of croplands and built-up lands (Cusens et al. 2024; Guo et al. 2023), the pressure of human activities is also still not negligible. The magnitude and distribution of watersheds and wetlands, which are the types of land use with high ESVs, is principally shaped by ecological forces. Sufficient precipitation directly determines the formation and maintenance of regional watersheds (Leung et al. 1996), which can enhance hydrological regulation and water supply of regional ecosystem. The topography also shape the flow pathways and distribution of watersheds (Leung et al. 1996). Coastal wetlands, as a unique land in coastal zone, have a pivotal role in governing water supply and hydrological regulation of coastal zone (Wei et al. 2024; Xu et al. 2020). The distribution of wetlands is equally dependent on precipitation and NDVI (Lin et al. 2019; Zhang et al. 2021). Abundant precipitation enhances soil moisture and promotes the growth of vegetation (Cui et al. 2022), both of which are essential for the long-term survival and expansion of wetlands (Taddeo et al. 2019). In contrast, the expansion of built-up lands and croplands has been largely attributed to socio-economic drivers, e.g., human activity intensity, gross domestic product and population density. The urbanization has accelerated the expansion of built-up lands (Liu et al. 2013, 2024b), especially in Dalian and Yingkou, weakening the magnitude of ESV in these regions. Moreover, the integrity of watersheds and wetlands has been destroyed by urban infrastructure construction and road networks, leading to habitat fragmentation and biodiversity deterioration, with the ultimate reduction of regional magnitude of ESV.

The land patterns and subsequent ESV in future would be driven by the policies and planning under divergent development scenarios (Asadolahi et al. 2018; de Moura et al. 2024; Guo et al. 2023). In the context of maximum economic benefits scenario, land planning generally prioritizes meeting the needs of economic growth and

urbanization. The demand for large amounts of built-up lands would continue to encroach on watersheds and wetlands for the purposes of urbanization, infrastructure construction, and industrial development. As a result, those specific lands (e.g., watersheds and wetlands) with high ESVs are expected to continue shrinking, with the decline of hydrological regulation, biodiversity, carbon sequestration and oxygen generation, despite economic growth achievable. In contrast, more attention is paid to the protection of specific lands with high ESVs in land planning that aims to emphasize ecological priority. The expansion of built-up lands would be strictly restricted, and the efforts to protect watersheds and wetlands would be prioritized. Ecological restoration practices in the context of ecological (maximize ecosystem service and/or ecological capacity) scenarios, e.g., the return of croplands to lakes and wetland restoration, are further promote the expansion of watersheds and wetlands (Carlucci et al. 2020; Fu et al. 2023). It is expected that those strategies enhance the magnitude of regional ESVs (Carlucci et al. 2020), e.g., hydrological regulation, climate regulation and biodiversity in coastal wetlands. The policy designs aiming at the balance of economic growth and ecological protection is essential to support the sustainable development of Liaoning coastal zone.

4.2 Coastal zone development suggestions

Considering the vital role of coastal zone in supporting economic growth and ensuring ecological security, those researches about the spatiotemporal patterns of coastal ESVs are of great theoretical and practical implications. This study identifies the spatiotemporal variations of ESV in Liaoning coastal zone from a perspective of “past-present-future”, with strong support for constructing the sustainable pattern of coastal protection (Bateman et al. 2013; Costanza 2020; Daily et al. 2009). The policymakers should, in view of long-term perspectives, consider local conditions for reasonable utilization of water and land resources (Bateman et al. 2013). To be specific, land-use policies should be implemented in a zoning-based manner considering local conditions. For ecologically sensitive areas and low-efficiency marginal croplands (e.g., Wafangdian), the government may continue ecological restoration measures (e.g., returning croplands to forests/lakes) to moderately increase vegetation coverage and the magnitude of ESVs (Bastin et al. 2019; Hua et al. 2016). In contrast, in key grain-producing areas (e.g., Zhuanghe and Dandong), cropland protection, consolidation, and productivity improvement should be prioritized to ensure food security. Moreover, the management of water and land resources in the regions of ‘Jinzhou-Panjin-Yingkou’ also should be strengthened to prevent the degradation of natural landscape (e.g.,

forests, wetlands and watersheds). The efficient utilization of water resources, like water-saving irrigation systems, is essential for improving water use efficiency and covering human water demands (Evans and Sadler 2008). The estimation of ESVs at grid scale can provide guidance for ecological protection planning and sustainable exploitation of coastal zone. Although the future scenarios in this study mainly emphasize human-driven land-use change, long-term coastal planning should also consider climate-related risks (e.g., sea-level rise and extreme events) when formulating restoration priorities and infrastructure layouts.

4.3 Limitations

This study also has some limitations. First, the classification of land use follows the first-level category of land because of the paucity of high-precision map along the broad coverage of Liaoning coastal zone. It is unable to precisely describe the features of internal structure and function in coastal regional ecosystem. More attentions should be devoted to the detailed classification of land use, like the subdivision of wetlands into coastal-, marsh- and artificial wetlands. Nevertheless, this work still provides a baseline of natural resources in Liaoning coastal zone, and opens opportunities to advance the understanding of how human activities may reshape the spatial patterns of coastal ESVs. Second, coastal ecosystem services generally are associated with multiple processes and/or parameters, and these processes and/or parameters exhibit high heterogeneities across various spatiotemporal scales (Ma et al. 2024; Yue et al. 2023), thus contributing to the uncertainties in ESV assessment (Yao et al. 2025). For instance, climate-related coastal processes and hazards—such as sea-level rise, storm surges, and extreme precipitation or drought—that not explicitly considered by this study, also may increasingly influence land-cover dynamics and ecosystem functioning, especially over longer time scales (Dee et al. 2025; Runting et al. 2017). Consequently, future studies should explore more integrated frameworks that couple human-demand-driven land-use scenarios with climate-related coastal processes to better capture the mechanisms underlying spatiotemporal fluctuations in regional ESVs (Baustert et al. 2018; Hamel and Bryant 2017). Moreover, the scenario analysis in this study was designed to explore trade-offs among multiple ecosystem service categories rather than to impose explicit minimum thresholds for individual provisioning services; consequently, outcomes under certain scenarios (e.g., a minor food-production deficit) should be interpreted as illustrative results of scenario-based trade-offs rather than as absolute management targets. In this context, minimum grain-production constraints were not explicitly imposed in the

optimization framework, but could be incorporated in future studies together with alternative pathways such as trade-based compensation or productivity improvement to further refine scenario design.

5 Conclusion

This study identifies the spatiotemporal variations of ESVs in Liaoning coastal zone from a ‘past-present-future’ perspective, providing an opportunity to improve the understanding of the interaction between land use change and ESVs. There was a notable expansion of built-up lands over the period 2000–2020 year, encroaching on considerable areas of croplands, watersheds and bare lands. The fluctuations of ESVs over past two decades are primarily in hydrological regulation (first increased by 3.36% and then decreased by 2.02%), biodiversity (first increased by 1.04% and then decreased by −9.14%) and landscape aesthetics (first increased by 1.03% and then decreased by −9.39%), which can be attributed to the changes of watersheds and wetlands. In 2050 year, the land patterns of Liaoning coastal zone under ecosystem service and ecological capacity scenarios are consistent in general, and the dominated land types are watersheds (518,665.90 hm²; 37.30%), built-up lands (326,634.37 hm²; 23.49%) and wetlands (233,399.47 hm²; 16.78%). The amount of ESVs in 2050 year in economic scenario (127.23 billion CNY) is higher compared to baseline scenario (89.14 billion CNY), but less than that of ecological scenarios (312.78 and 311.48 billion CNY for ecosystem service and ecological capacity scenarios, respectively). The spatiotemporal patterns of coastal ESV are constrained by both anthropogenic and natural drivers. Therefore, in the process of coastal development, policymakers need to focus on the contribution of anthropogenic and natural factors on land expansion, benchmarking divergent development targets and regional natural resources, so as to achieve the coordination between land optimization and economic development.

Supplementary Information

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Supplementary Material 1

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Authors' contributions

Yanzhong Yao and Youtao Song conceived this research. Yanzhong Yao, Junru Zhou and Changgeng Jia collected the data. Yanzhong Yao analyzed the data.

Yanzhong Yao wrote the draft. Yanzhong Yao and Pinjie Su revised the manuscript. All authors contributed to the reviewing of the manuscript.

Data availability

Data will be made available on request.

Declarations

Competing interests

The authors declare that there is no conflict of interest.

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